Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	TE-AI&DS	Semester:	
Course Code:	2015118	Course Name:	Artificial Intelligence & Soft Computing Lab

Name of Student:	Rushikesh Subhash Khemnar
Roll No. :	81
Experiment No.:	8
Title of the Experiment:	Development of Small Language Models (SLMs) and Micro Language Models
Date of Performance:	21/09/2026
Date of Submission:	28/09/2026

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timel submission	10	
Total	20	

Performance Indicator	Exceed Expectations EE	Meet Expectations	Below Expectations BE
Performance	4-5	2-3	1
Understandin	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty**:**

Signature

Date

Aim: To develop Small Language Models (SLMs) and Micro Language Models (gLMs).

Objectives:

After completing this experiment, the student will be able to:

1. To understand the concepts of Small Language Models (SLMs) and Micro Language Models (YLMs).
2. To explore the architecture and working of lightweight language models.
3. To develop and execute a basic Small Language Model for text-based tasks.
4. To analyze the performance of SLMs and gLMs for simple language processing applications.
5. To understand the advantages and limitations of lightweight language models..

Tools Required:

- Hardware:
- Desktop/Laptop o Internet Connection
- Software: o Python 3 .x o Visual Studio Code / Jupyter Notebook o Hugging Face Transformers Library o Ollama / Google Colab (Optional) o Windows/Linux Operating System

Dataset Details:

A small text dataset or publicly available sample dataset may be used for developing and testing the language model. Alternatively, pre-trained Small Language Models (SLMs) available through open-source frameworks may be utilized for experimentation.

Theory:

Small Language Models (SLMs) and Micro Language Models (gLMs) are compact versions of Large Language Models designed to perform natural language processing tasks with lower computational requirements. These models require less memory, faster inference time and lower power consumption, making them suitable for deployment on edge devices, mobile applications and resource-constrained systems.

SLMs and gLMs are capable of performing tasks such as text generation, text classification, summarization, and question answering while maintaining acceptable performance for domain-specific applications. Their lightweight architecture enables efficient execution without requiring high-end computing resources, making them an important component of modern Artificial Intelligence applications.

Procedure:

1. Install the required libraries and language model framework.
2. Select a suitable Small Language Model (SLM) or Micro Language Model (ULM).

3. Configure the development environment.
4. Load the selected language model.
5. Provide suitable text input or prompts.
6. Execute the model and observe the generated output.
7. Analyze the performance and record the observations.

Description of the Experiment:

This experiment focuses on the development and implementation of Small Language Models (SLMs) and Micro Language Models (PLMs) using open-source tools and frameworks. A lightweight language model is configured and executed to perform basic natural language processing tasks such as text generation or text completion.

The experiment provides an understanding of the workflow involved in developing compact language models and demonstrates their practical applications in environments with limited computational resources.

Detailed Description:

Small Language Models (SLMs) and Micro Language Models (gLMs) are designed to provide efficient language processing capabilities while minimizing computational complexity. In this experiment, a lightweight language model is developed or deployed using an open-source framework and evaluated for basic natural language processing tasks. Appropriate text prompts are provided to the model and the generated responses are observed to understand its language generation capability.

The performance of the selected model is evaluated based on the quality and relevance of the generated output, execution speed and resource utilization. The experiment highlights the suitability of lightweight language models for applications requiring faster inference, reduced memory consumption and efficient deployment on edge devices and embedded systems.

Conclusion:

Small Language Models (SLMs) and Micro Language Models (gLMs) were successfully developed and evaluated for basic natural language processing tasks. The experiment demonstrated the effectiveness of lightweight language models in providing efficient language processing with reduced computational requirements.

Code:

```
from transformers import pipeline import
time print("Loading Small Language
Model...")

generator = pipeline( "text-generation",
   model="distilgpt2"
```

```
prompt = "what is Artificial Intelligence?" start

time time.time()

result generator( prompt, max
   new tokens=50, num return
   sequences=l,                do
   sample=True,
   temperature=0.7
)

end_time time.time()

print("\nlnput Prompt: ") print(prompt)

print("\nGenerated Output:" ) print(result[()]["generated text"])
print("\nExecution Time: {:.2f} seconds" .format(end time - start time))
```

Output:

```
model .sdfetensors; dwnLoeding bytes;                                                    B. 57*3/s
model safetensors•: reconstructing file.

Loading weights: lee                                                  76/76              163.a1if,'5]

generation_confie.json: 1Ø3%l                                         | 124,'124

tokenizer_config.json;                                                | 26.0/26.8
vocab merges .txt:          ] 4560456k tokenizer.json:

[transformers] Passing generetion_config- together with generation-related , ' tenpereture' •
is deprecated and will be ram.'ed in future versions. Please pass either a '#'nerafion config' object OR all generation par³meters explicitly; buf not both. [transformers] 30th (—
9B) and (—50) to nave been set. will take precedence. Please refer to the documentatio n -for more information.
(https:%'huggingface.co/docs/transformers/main'en/main_classes'text_generation)
[transformers] Ignoring for 3PE tokenizer• CPT2Tokenizer. The post-processing ste. is designed for %rdP iece tokenizers and is destructive for BRE (it strips spaces before
punctuation). Set to suppress this warning, or set clean to force cleanup anymay.

Input Prmgt:
Artificial Intelligence
Generated Output:
Artificial Intelligence is e        fom of deta visualization. It has •ISO beccne dn -important tool to develop  predictive models for the future.


The main            of the visualization -is to detonstrate hm the world is changing.

In Sepf&ber 2816
```

Questions to be Answered by Students:

1. Which Small Language Model (SLM) or Micro Language Model (gLM) was used in the experiment?

Answer: The DistilGPT-2 model was used. It is a lightweight, pre-trained language model available through the Hugging Face Transformers library.

2. What type of text output was generated by the language model?

Answer: The model generated text continuation/text generation output based on the given prompt. For example, from the prompt "Artificial Intelligence is it generated related sentences automatically.

3. What are the advantages of using Small Language Models over Large Language Models?
   Answer:

- Require less memory and computational power.
- Provide faster inference and response time.
- Consume less power.
- Can run on ordinary computers and edge devices.
- Are easier and less expensive to deploy.
- Suitable for simple and domain-specific NLP tasks.

4. State two observations obtained from the implementation of the Small Language Model (SLM).

Answer:

l. The SLM successfully generated meaningful text based on the input prompt.

2. The model required relatively low computational resources and produced the output within a short execution time compared with larger language models.